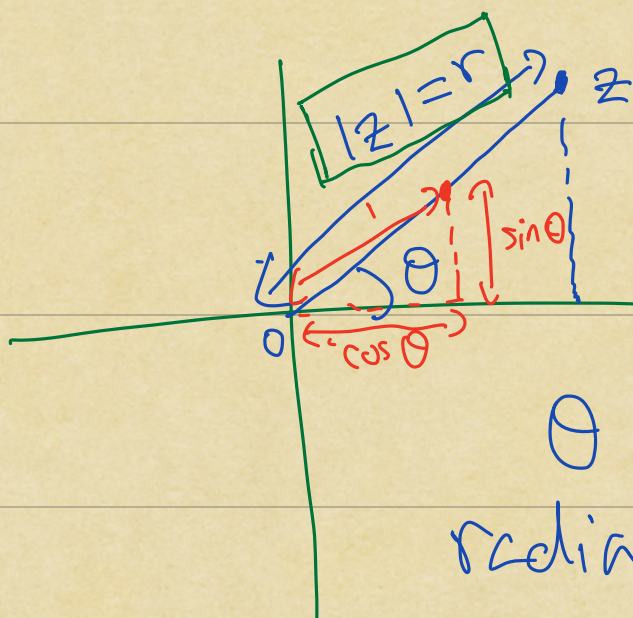


Complex numbers

Euler's notation



$$z = r e^{i\theta} \\ = r e^{i\theta}$$

θ measured in radians, $\theta \in [0, 2\pi)$

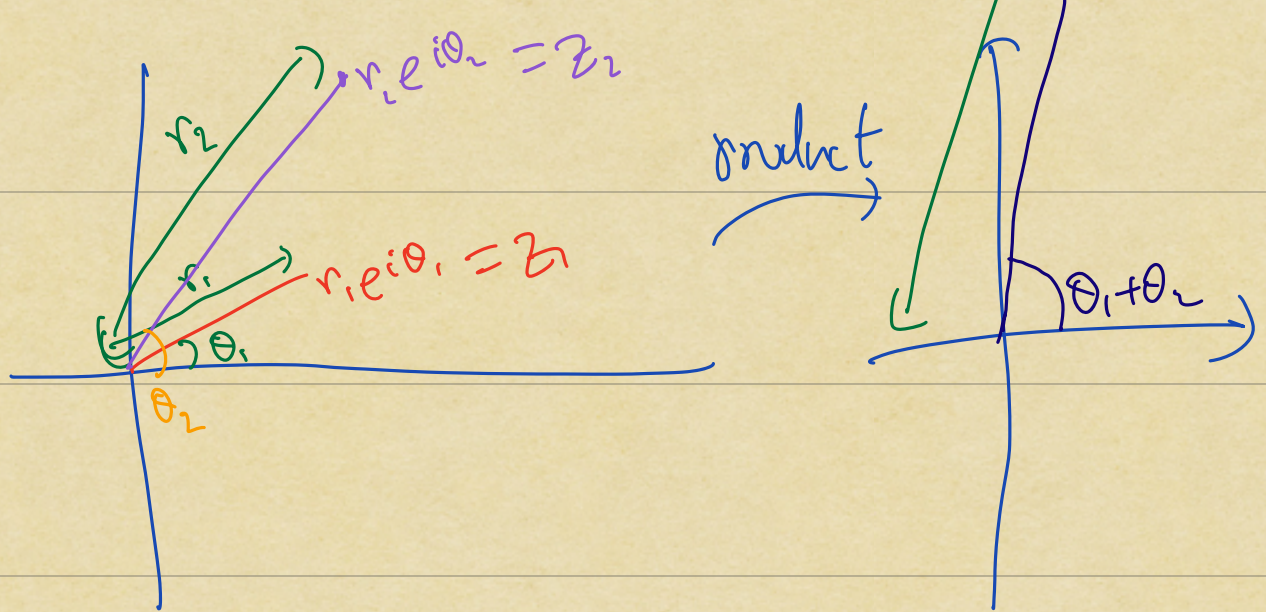
Euler's formula

$$e^{i\theta} = \cos \theta + i \sin \theta$$

Trig formulas $\Leftrightarrow$

$$e^{i\theta_1} \cdot e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$$

Diagram illustrating the multiplication of two complex numbers in polar form. Two vectors are shown: one with magnitude r_1 and angle θ_1 , and another with magnitude r_2 and angle θ_2 . Their product is a vector with magnitude $r_1 r_2$ and angle $\theta_1 + \theta_2$.



Example:

Solve $z^3 = 1$ in $\mathbb{C}$

Write $z = r e^{i\theta}$

Then we need to solve

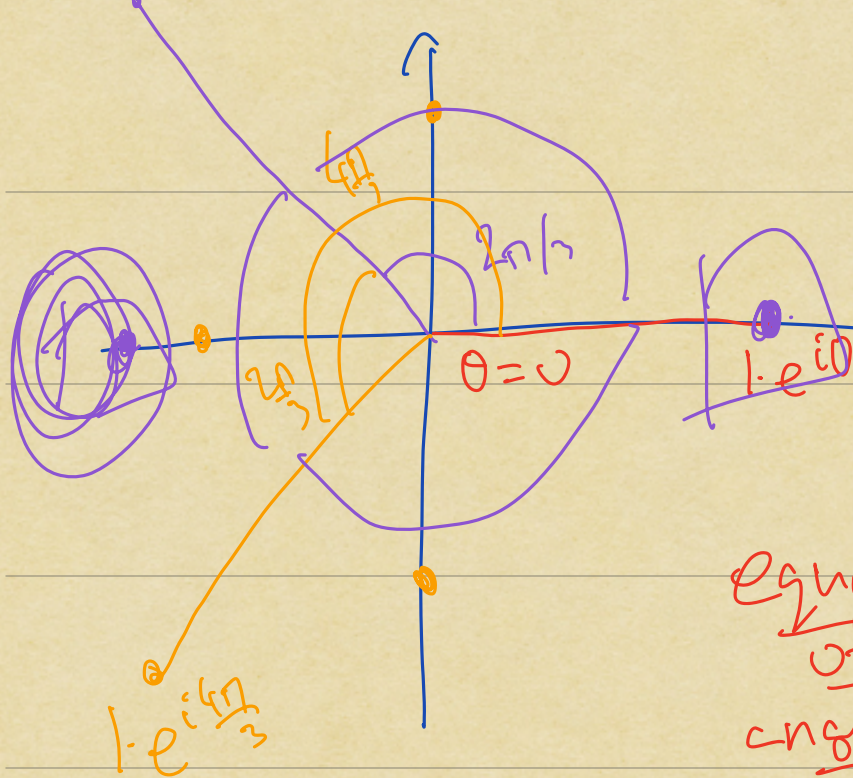
$$(r e^{i\theta})^3 = 1$$



$$r^3 e^{i3\theta} = r^3 (e^{i\theta})^3 = 1$$

$$\Rightarrow r^3 = 1 \Leftrightarrow r = 1$$

$$1 \cdot e^{i\frac{2\pi}{3}}$$



$$0 \cdot 3 = 0$$

$$\frac{2\pi}{3} \cdot 3 = 2\pi \quad \frac{\omega_s}{\text{angle}} \quad 0$$

$$\frac{4\pi}{3} \cdot 3 = 4\pi \quad \frac{\omega_s}{\text{angle}} \quad 0$$

Note:

$$\boxed{\begin{aligned} \frac{6\pi}{3} &= 2\pi = 0 \\ \frac{8\pi}{3} &= \frac{2\pi}{3} \end{aligned}}$$

equalities
of
angles

Rmb: The same technique
applies for the equation

$$z^n = \alpha$$

for fixed $\alpha \in \mathbb{C}$
 $\alpha \neq 0$

There are n solutions

$$1 \neq \alpha = se^{i\theta}$$

then the solutions are

$$z = s^{1/n} e^{i(\theta + 2\pi k)/n}$$

$$k = 0, 1, \dots, n-1$$

Takeaways

Examples of fields

Finite

$$\mathbb{Z}/p\mathbb{Z}$$

p prime

Infinite

$$\mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

↑
rational

↑
real

↑
Complex

Thm (Galois)

The only possible sizes
of finite fields are
powers of primes p^d

Moreover, for every such
power, there is a unique
field of size p^d .

Example:

F : field

$a, b, c \in F$; then we can
write down a

polynomial $\boxed{a}x^2 + \boxed{b}x + \boxed{c} = f(x)$

Can we find $f(x)$ s.t.

$$f(1) = 1 \Leftrightarrow a \cdot \underline{1}^2 + b \cdot \underline{1} + c = 1$$

$$f(-1) = -1 \Leftrightarrow a \cdot \underline{(-1)}^2 + b \cdot \underline{(-1)} + c = -1$$

$$f(0) = 1 + 1 = 2 \neq 0,$$

$$\Leftrightarrow a \cdot 0 + b \cdot 0 + c = 2$$

a, b, c are our variables
for a linear system

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$\downarrow R_2 - R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & -2 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$\downarrow R_2 \times (-2)^{-1}$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

This
would fail
if $-2 = 0 \in \mathbb{F}$

$$\downarrow R_1 - R_2$$

e.g. if
 $F = 2/27$

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$\downarrow R_1 - R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Solution

$$f(x) = -2x^2 + x + 2$$

Rmk: What is

happening here is that

the space of polynomials
of $\deg \leq 2$ is a

vector space

over F , and we are
solving a linear problem
in this vector space

Vector Space

F : field

Defⁿ A vector space

V over F is a

set V equipped with:

(Zero) (i) $0 \in V$ [$0_V \in V$ ^{also}]

(Addition) (ii) $+$: $V \times V \rightarrow V$
 $(v_1, v_2) \mapsto v_1 + v_2$

(Scalar
multiplication)

(c)

Scalars

$$\bullet : \boxed{F} \times V \rightarrow V$$
$$(a, v) \mapsto a \cdot v$$

Satisfying :

(1)

$$0 + v = v + 0 = v$$

(Additive
identity)

$$\forall v \in V$$

redundant : $-v = (-1) \cdot v$

(2)

$$\forall v \in V, \exists -v \in V$$

s.t.

$$v + (-v) = 0$$

(Additive
inverse)

(3)

$$v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$$

(Associativity)

$$\forall v_1, v_2, v_3 \in V$$

(4)

$$v_1 + v_2 = v_2 + v_1$$

(Commutativity)

$$\forall v_1, v_2 \in V$$

(5)

$$a \cdot (v_1 + v_2)$$

$$= a \cdot v_1 + a \cdot v_2$$

(Distributivity)

$$\forall a \in F$$

$$v_1, v_2 \in V$$

(6)

$$a_1 \cdot (a_2 \cdot v) = (a_1 a_2) \cdot v$$

(Associativity
for scalar
multiplication)

$$\forall a_1, a_2 \in F$$

$$v \in V$$

(7)

$$\forall v \in V, 1 \cdot v = v$$

(Identity)

Handwritten notes in green ink at the top left corner.

Blank lined paper with horizontal ruling lines.